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Chapter 1. What is reinforcement learning?

Figure 1.1. An image classifier is a function or learning algorithm that takes in an image and returns a class label, classifying the image into one of a finite number of possible categories or classes.

Figure 1.2. Perhaps the simplest machine learning model is a simple linear function of the form f(x) = mx + b, with parameters m (the slope) and b (the intercept). Since it has adjustable parameters, we call it a parametric function or model. If we have some 2-dimensional data, we can start with a randomly initialized set of parameters, such as [m = 3.4, b = 0.3], and then use a training algorithm to optimize the parameters to fit the training data, in which case the optimal set of parameters is close to [m = 2, b = 1].

Figure 1.3. A sentence like “John hit the ball” can be decomposed into simpler and simpler parts until we get the individual words. In this case, we can decompose the sentence (denoted S) into a subject noun (N) and a verb phrase (VP). The VP can be further decomposed into a verb, “hit,” and a noun phrase (NP). The NP can then be decomposed into the individual words “the” and “ball.”

Figure 1.4. As opposed to an image classifier, a reinforcement learning algorithm dynamically interacts with data. It continually consumes data and decides what actions to take—actions that will change the subsequent data presented to it. A video game screen might be input data for an RL algorithm, which then decides which action to take using the game controller, and this causes the game to update (e.g. the player moves or fires a weapon).

Figure 1.5. Deep learning is a subfield of machine learning. Deep learning algorithms can be used to power RL approaches to solving control tasks.

Figure 1.6. This graphical depiction of words in a 2D space shows each word as a colored point. Similar words cluster together, and dissimilar words are farther apart. Data naturally lives in some kind of “space”